Microeconomics **Problem Set 6**, Spring 2023

1. (Externality) Clyde produces chemicals. His cost function is $C_c=5c^2+100c$, where c is the number of liters of chemicals produced. The market price for a liter of a chemical is $p_c=700$. Bonnie is a baker. She is adversely affected by the noxious fumes emitted by Clyde's production of chemicals. Her cost function is $C_b(b,c)=0.5b^2-140b+bc$, where b is the number of baked goods produced. The market price for a baked good is $p_b=10$.

   a. How many liters of chemicals and how many baked goods will be produced in the market equilibrium?

$$p_c = MC_c$$

$$700 = 10c + 100$$

$$c^M = 60,$$

$$p_b = MC_b$$

$$10 = b - 140 + c$$

$$b^M = 90.$$

   b. Suppose Clyde and Bonnie were to jointly maximize profits. How many liters of chemicals and how many baked goods will be produced?

$$\max_{c,b} \pi = \pi_c + \pi_b$$

$$= \left[700c - \left(5c^2 + 100c\right)\right] + \left[10b - \left(\frac{1}{2}b^2 - 140b + bc\right)\right]$$

$$\frac{\partial \pi}{\partial c} = 700 - 10c - 100 - b = 0$$

$$b = 600 - 10c$$

$$\frac{\partial \pi}{\partial b} = 10 - b + 140 - c = 0$$

$$b = 150 - c.$$

1. $C_c = 5c^2 + 100c \quad P_c = 700$

$C_b(b,c) = 0.5b^2 - 140b + bc \quad P_b = 10$

a. $P_c = MC_c$

$\Rightarrow 700 = 10c + 100 \Rightarrow c = 60$

$P_b = MC_b$

$\Rightarrow 10 = b - 140 + c \quad , \quad c = 60$

$10 = b - 140 + 60 \Rightarrow b = 90$

b. $\max \pi = P_c \cdot c + P_b \cdot b - (5c^2 + 100c) - (0.5b^2 - 140b + bc)$

$= 700c + 10b - 5c^2 - 100c - 0.5b^2 + 140b - bc$

$[c] = 700 - 10c - 100 - b = 0 \Rightarrow 600 = 10c + b$

$[b] = 10 - b + 140 - c = 0 \Rightarrow 150 = b + c$

$\Rightarrow 450 = 9c \Rightarrow c = 50 \quad b = 100$

c. joint profit $= 700 \times 50 + 10 \times 100 - 5 \times 50^2 - 100 \times 50$

$- 0.5 \times 100^2 + 140 \times 100 + 50 \times 100$

$= 22500$

d. Clyde pays the tax

marginal tax rate = marginal externality cost

$C_b(C,b) = 0.5b^2 - 140b + bC$

$\frac{\partial C_b(c,b)}{\partial c} = b$ at the efficient allocation

$b^* = 100$ tC

$\Rightarrow C_b(b,c) = 0.5b^2 - 140b + 100c$ $t = 100$

e. from c^M to c^* $\Rightarrow 60 \rightarrow 50$

which Clyde needs to pay (22,500 − 22,050) = 450 to Bonnie.

Solving equations (12) and (13) simultaneously, we find that $c^* = 50$ and $b^* = 100$.

c. Compare the competitive equilibrium outcome from part a and the jointly profit-maximization outcome from (b). Which outcome is Pareto optimal? Why?

Competitive market outcome: $(c^M, b^M) = (60, 90)$

$$\pi^M = \pi_c + \pi_b$$

$$= \left[700 \cdot 60 - \left(5 \cdot 60^2 + 100 \cdot 60\right)\right] + \left[10 \cdot 90 - \left(\frac{1}{2} \cdot 90^2 - 140 \cdot 90 + 90 \cdot 60\right)\right]$$

$$= 18,000 + 4,050$$

$$= 22,050.$$

Joint profit-maximization outcome: $(c^*, b^*) = (50, 100)$

$$\pi^* = \pi_c + \pi_b$$

$$= \left[700 \cdot 50 - \left(5 \cdot 50^2 + 100 \cdot 50\right)\right] + \left[10 \cdot 100 - \left(\frac{1}{2} \cdot 100^2 - 140 \cdot 100 + 100 \cdot 50\right)\right]$$

$$= 17,500 + 5,000$$

$$= 22,500.$$

Total profits from joint profit maximization, π^*, are greater than total profits from individual profit maximization, π^M. If Bonnie transfers a minimum of \$500 and a maximum of \$950 to Clyde, then $\pi_c^* \geq \pi_c^M$ and $\pi_b^* \geq \pi_b^M$. Thus, $(c^*, b^*) = (50, 100)$ is Pareto optimal.

d. One of the solutions to the externality problem is corrective taxes. In this example, who pays the tax? How does the tax solve the externality problem? What is the marginal tax rate?

Since this is a negative externality, the creator of the externality, Clyde, pays the tax. The negative externality imposed on Bonnie by Clyde is now internalized by Clyde. The marginal tax rate to restore efficiency should be the marginal external cost, which was ignored by Clyde in his individual maximization. That is, $\partial C_b/\partial c = b$. This, at the efficient allocation, is $b^* = 100$. Thus, the Pigouvian solution relies on having Clyde maximize his original individual profits minus tc, where $t = 100$.

e. Another solution is Coasian property rights. Suppose Clyde has the legal right to emit noxious fumes. How can they arrive at the Pareto optimal outcome?

If Clyde and Bonnie are able to negotiate a mutually beneficial contract where Bonnie pays Clyde to reduce his output from c^M to c^*, then they will end up at the Pareto optimal outcome, (c^*, b^*). The size of the transfer payment was already identified at [redacted] part c.

2. (Public Goods) Seven dwarfs live in a little house in the forest. The Public Good Board, which calls itself "Snow White," has discovered that the dwarf's hands are dirty all the time and is proposing a policy to solve the problem. The central part of the policy is to purchase a sink, and have it installed in the bathroom of the dwarf's house. Let $x \geq 0$ represent the size of the sink. Forest econometricians have determined that the cost of buying and installing a sink of size x is given by the function $C(x)=x^2/112^{0.5}$. For the seven dwarfs living in the house, the sink is a public good. Suppose dwarf i's utility function is $u_i(x, y_i)=x^{0.5}+y_i$, where y_i is the consumption of all private goods, measured in dollars. Dwarf I's initial endowment of the private good -money- is $\omega_i>0$.

a. Using the Samuelson optimality condition to find the efficient size for the sink.

The Samuelson condition for optimality of the public good requires that the sum of the marginal rates of substitution for each dwarf equates the marginal cost of producing the public good. That is,

$$7 \times \frac{1}{2\sqrt{x}} = 2x/\sqrt{112},$$

which reduces to

$$(7/4) \times \frac{1}{\sqrt{x}} = x/\sqrt{112}.$$

This yields $x^* = 7$.

2. $C(x) = \frac{x^2}{\sqrt{112}}$ $\qquad w_i > 0$

$u_i(x_i, y_i) = x^{0.5} + y_i$

a. $mB_1(x) + \cdots + mB_7(x) = mC(x)$

$$\Rightarrow 7 \times \frac{1}{2} \cdot \frac{1}{\sqrt{x}} = \frac{2x}{\sqrt{112}}$$

$$\Rightarrow \frac{7}{4} \cdot \frac{1}{\sqrt{x}} = \frac{x}{\sqrt{112}} \Rightarrow \frac{7}{\cancel{4}} \cdot \frac{1}{\sqrt{x}} = \frac{x}{\cancel{4}\sqrt{7}} \Rightarrow x = 7$$

b. $\max u_1(x, y_1) = x^{0.5} + y_1$

s.t. $y_1 + \frac{x^2}{\sqrt{112}} = w_1$

$$[x] = \frac{1}{2\sqrt{x}} - \frac{2x}{\sqrt{112}} = 0 \Rightarrow x_i = 7^{\frac{1}{3}}$$

$$\max u_j = \frac{1}{2\sqrt{x_i}} < 1, \quad x_i = 7^{\frac{1}{3}}$$

$$\Rightarrow x_i = 7^{\frac{1}{3}}, \quad x_j = 0 \quad \forall\, i \neq j$$

at market eqm. all dwarves but one free-rider on dwarf i and the size of the sink purchased is much smaller than the efficient size.

b. What would happen if the sink were privately purchased by the dwarfs? (Assume that dwarf 1 would first arrange to buy and pay for some sink, and then dwarf 2 would arrange to buy and pay for some additional sink, and so on. The buying process would stop when no dwarf wants to add any more sink. Then the sink that they had arranged to buy would finally be set up in their bathroom.) Determine the ultimate size of the purchased sink, and indicate who pays what toward its purchase.

In order to figure out how much of the public good he would like to purchase in the private market, each dwarf solves the problem of maximizing his utility function subject to the budget constraint $y_i + x_i^2/\sqrt{112} = \omega_i$. This yields a first-order condition:

$$\frac{1}{2\sqrt{x}} = \frac{2x_i}{\sqrt{112}}.$$

The solution to this equation is $x_i = 7^{1/3}$ and $x_j = 0$ for all $j \neq i$. Such a profile also solves the FOC of the utility maximization problems for all dwarves $j \neq i$. That is, at the market equilibrium, all dwarves but one free-ride on dwarf i and the size of the sink purchased is much smaller than the efficient size.